new tools and techniques for composing functions and programs and new ways of thinking about how to design them would be needed. For now, though, you have enough to start writing applications, from functions all the way down.

3.8. Summary

Functions are first-class values in CoffeeScript. They're used for structuring programs at several levels and allow you to group together multiple expressions, treating them like a single expression. By assigning function values to a variable, you have names for functions that you use to organize programs.

First-class functions used as callbacks are a fundamental I/O technique used both in the browser and on the client. You've seen repeatedly how this technique works in different contexts and how it allows I/O to be treated with the same techniques whether it's local file access or across a network. This idea has some important implications and advantages that will be explored later on.

Functions are combined by invoking them inside other functions, by passing them as arguments to other functions, and by using functions as return values from other functions. By gluing functions together, you can use them to create abstractions of other functions, as you saw in the `accumulator` function.

Closure provides a way to isolate parts of a program by containing the variables defined within them, and by passing closures around in your programs you can use them to carry data. In one sense, closures provide some of the features that you'll see objects provide in the next chapter but with a different conceptual model and a different sweet spot for where they're used.

Finally, CoffeeScript blends ideas from different programming languages. Although it supports some important functional programming techniques, it